H

VICTORIA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME			
CLASS		TUTOR'S NAME	

PHYSICS 9478/02

Paper 2 Structured Questions **2 September 2026**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and tutor's name in the spaces at the top of this page.

Write in dark blue or black pen on both sides of the paper.

You may use a HB pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 9
2	/ 8
3	/ 9
4	/ 10
5	/ 10
6	/ 9
7	/ 20
Total	/ 75

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} $(1 / (36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = F / A$
gravitational potential	$\phi = -Gm / r$
temperature	$T / \mathrm{K} = T / ^\circ\mathrm{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal molecule	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
capacitors in series	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2} QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2} CV^2$
charging a capacitor	$Q = Q_0 \left[1 - e^{-\frac{t}{\tau}} \right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2} n^2$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions for a system to be in static equilibrium.

Vector sum of forces acting on the system = 0 [1]

Sum of clockwise moments about a pivot = sum of anticlockwise moments about the same pivot. [1]

[2]

(b) A uniform wooden rod of length 0.30 m and mass 0.45 kg is attached to a vertical wall with a hinge at an angle of 40° with the horizontal. The other end of the rod is attached to the wall with a rope, so that the rope makes an angle of 30° with the horizontal. An object M of mass 1.2 kg is attached to the other end of the rod, as shown in Fig. 1.1 below.

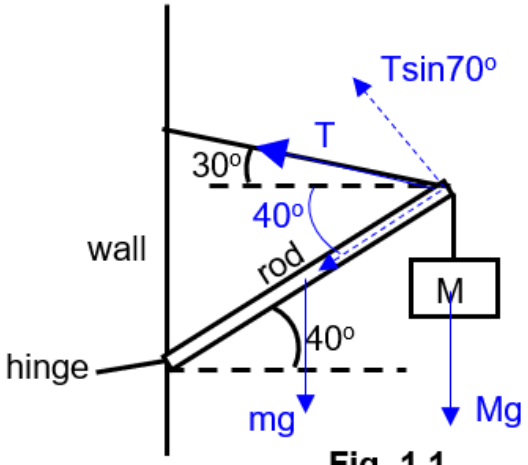


Fig. 1.1

(i) Show that the tension in the rope is 11 N.

Let length of rod be L.
Take moments about hinge.
Anticlockwise moment = clockwise moment

$$T\sin 70^\circ \times L = \left(Mg \times L\cos 40^\circ\right) + \left(mg \times \frac{L}{2}\cos 40^\circ\right)$$

$$T = \frac{\left(M + \frac{m}{2}\right) g\cos 40^\circ}{\sin 70^\circ}$$

$$= \frac{\left(1.2 + \frac{0.45}{2}\right) \times 9.81 \times \cos 40^\circ}{\sin 70^\circ}$$

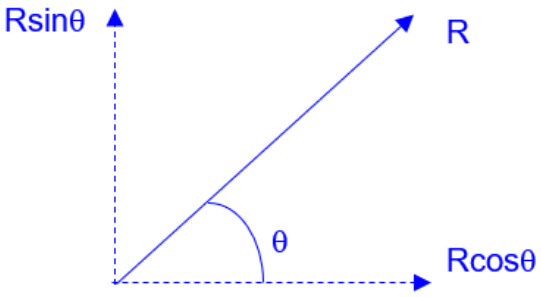
$$= 11.395$$

$$\approx 11\ \text{N}$$

[1 for correct anticlockwise moment; 1 for correct total clockwise moment
1 for more sf before final answer]

[3]

(ii) Determine the magnitude of the reaction force from the hinge acting on the rod, and the angle that the reaction force makes with the horizontal.



Take upwards and rightwards to be positive.

Let reaction force be R

Horizontally

$$R\cos\theta = T\cos 30^\circ$$
$$= 11\cos 30^\circ$$
$$= 9.526 \text{ N} \quad (1) \quad [1 \text{ mark}]$$

Vertically

$$R\sin\theta + T\sin 30^\circ = (M+m)g$$
$$R\sin\theta = (M+m)g - T\sin 30^\circ$$
$$= \left[(1.2+0.45)\times 9.81\right] - 11\sin 30^\circ$$
$$= 10.69 \text{ N} \quad (2) \quad [1 \text{ mark}]$$

$(2)\div(1)$ $\quad \tan\theta = \dfrac{10.69}{9.526}$

$$\theta = 48.30^\circ$$
$$\approx 48^\circ \quad [1 \text{ mark}]$$

Sub into (1) $\quad R\cos 48.30^\circ = 9.526$

$$R = 14 \text{ N} \quad [1 \text{ mark}]$$

magnitude = ... N

direction = ... ° from the horizontal

[4]

[Total: 9]

2 A speedboat of mass 450 kg is equipped with two identical motors. Each motor can produce 75 kW of power. When both motors are turned on, its maximum speed is 30 m s^{-1}. Assume that the drag force on the speedboat is proportional to the square of its speed.

(a) Determine the maximum speed of the boat when only one motor is turned on.

At maximum speed, driving force from motor = resistive force F.
Also $F = kv^2$, where k = constant.

Power supplied by motor $P = Fv_{max}$

$$= \left(kv_{max}^{\ 2}\right)v_{max}$$

$$= kv_{max}^{\ 3} \qquad [1]$$

$$\frac{P_1}{P_2} = \left(\frac{v_{max1}}{v_{max2}}\right)^3$$

$$v_{max1} = v_{max2} \times \sqrt[3]{\frac{P_1}{P_2}}$$

$$= 30 \times \sqrt[3]{0.5}$$

$$= 23.811$$

$$\approx 24 \text{ m s}^{-1} \qquad [1]$$

maximum speed = ... m s^{-1} [3]

(b) When both motors are turned on, the speedboat takes 9.0 s to reach the maximum speed of 30 m s^{-1}. Determine the amount of energy that is lost to the surroundings during that time.

Energy supplied by motor - energy lost = ke gained [1]

$$P_2 t - E_{lost} = \frac{mv^2}{2} - 0$$

$$E_{lost} = P_2 t - \frac{mv^2}{2}$$

$$= \left(150000 \times 9.0\right) - \frac{450 \times 30^2}{2}$$

$$= 1.1475 \times 10^6 \text{ J}$$

$$\approx 1.1 \times 10^6 \text{ J} \qquad [1]$$

amount of energy lost = ... J [2]

(c) Sketch, on Fig. 2.1, the variation with time t of the speed of the speedboat v, until it reaches maximum speed.

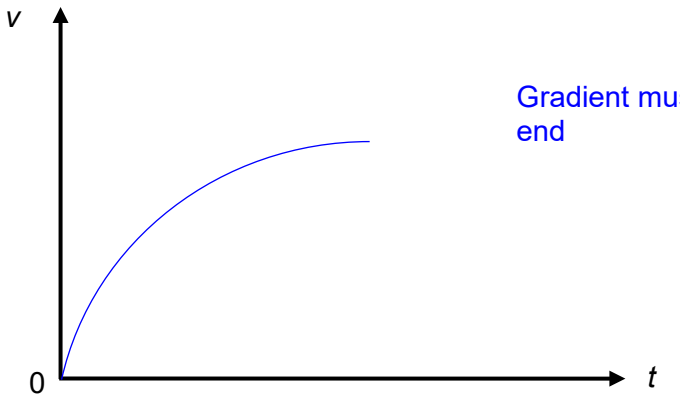


Fig. 2.1

[1]

(d) When the speedboat is travelling at a speed of 20 m s^{-1}, both motors are turned off at the same time. The speedboat continues to travel an additional distance of 600 m before coming to rest. Determine the average resistive force acting on the speedboat.

Work done against resistive force = ke lost [1]

$$fx = \frac{mv^2}{2} - 0$$

$$\text{Average resistive force } f = \frac{mv^2}{2x}$$

$$= \frac{450 \times 20^2}{2 \times 600}$$

$$= 150 \text{ N} \quad [1]$$

average resistive force = ... N [2]

[Total: 8]

3 **(a)** A small ball of mass 10 g rests at point P on a curved track of radius *r*, as shown in Fig. 3.1.

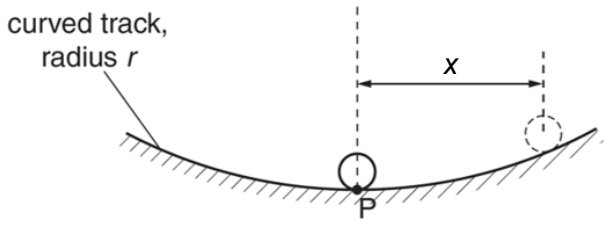


Fig. 3.1

The ball is moved a small distance to one side and is then released. The displacement *x* of the ball is related to its acceleration *a* towards P by the expression

$$a = -\frac{gx}{r}$$

where *g* is the acceleration of free fall.

(i) Explain how the above expression shows that the ball undergoes simple harmonic motion.

Since g/r is constant, a is proportional to x. [1]

The minus sign shows that a is opposite to x. [1]

[2]

(ii) The radius *r* of curvature of the track is 28 cm.
Determine the time interval τ between the ball passing point P and its first return to point P.

Compare given equation with $a = -\omega^2 x$,

$\omega^2 = \frac{g}{r}$ [1]

$(\frac{2\pi}{T})^2 = \frac{9.81}{28 \times 10^{-2}}$

T = 1.06 s [1]

Time taken between ball passing point P and its first return to point P

= $T/2$ = 1.06/2 = 0.53 s [1]

τ = ... s [3]

(b) The variation with time t of the displacement x of the ball in **(a)** is shown in Fig. 3.2.

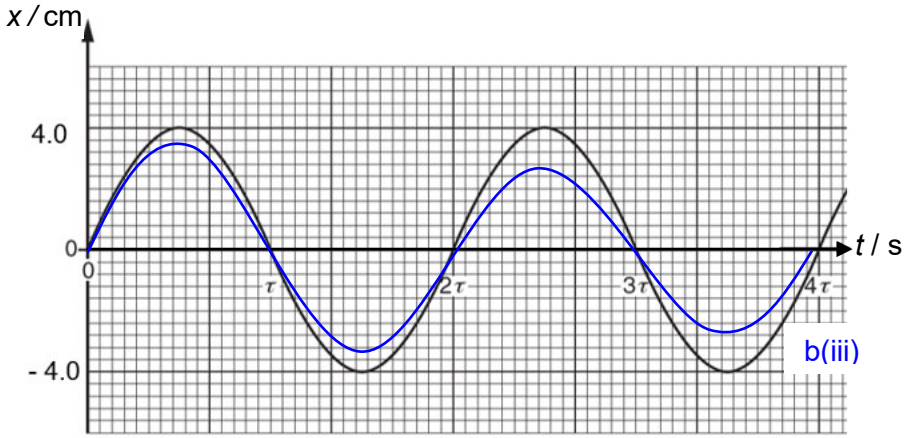


Fig. 3.2

(i) Calculate the maximum kinetic energy of the ball.

$$KE_{max} = \frac{1}{2} m\omega^2 x_o^{\ 2}$$

$$KE_{max} = \frac{1}{2}(10\times10^{-3})(\frac{9.81}{28\times10^{-2}})(4.0\times10^{-2})^2 \text{ [1]}$$

$$KE_{max} = 2.8\times10^{-4} \text{ J [1]}$$

maximum kinetic energy= .....J [2]

(ii) On Fig. 3.3, sketch the variation with time t of kinetic energy E of the ball for the first two periods. Include a suitable scale on the energy axis.

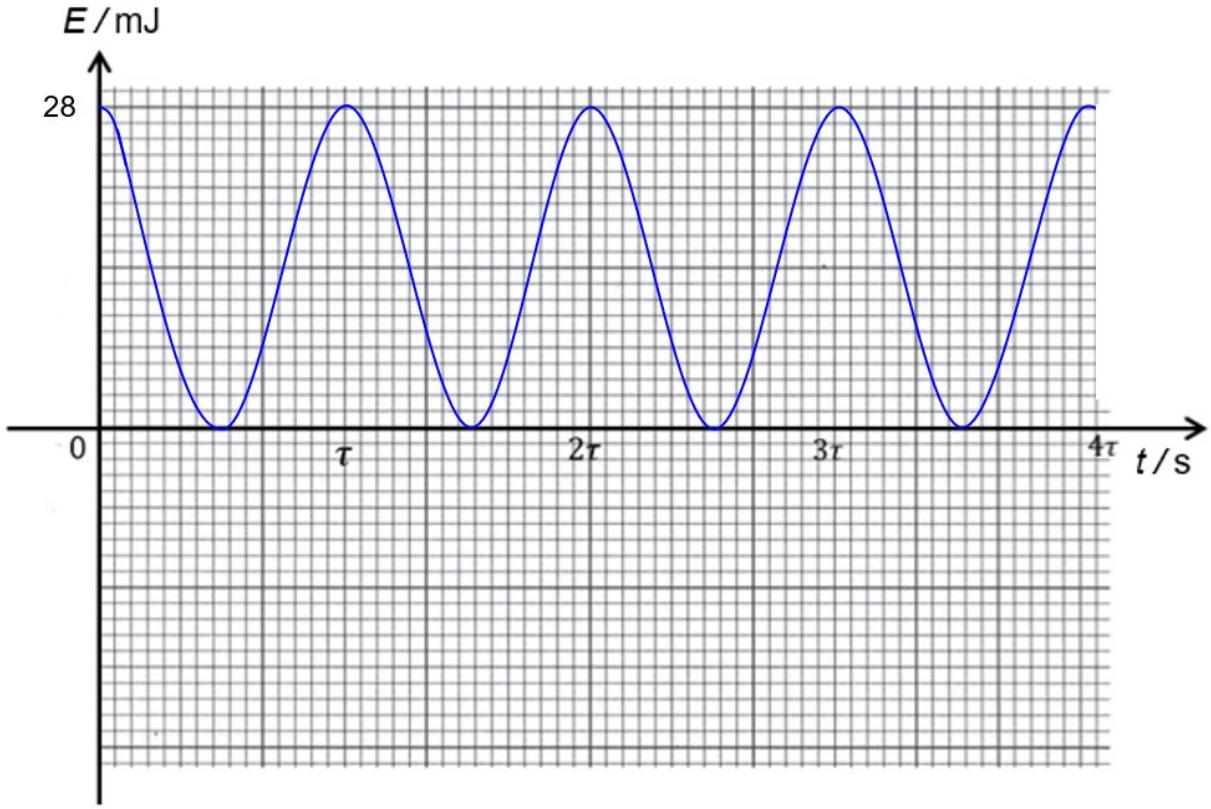


Fig. 3.3

[1]

(iii) Some moisture now forms on the track, causing the ball to come to rest after approximately after 15 oscillations.

On the axes of Fig. 3.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. [1]

Amplitude drops with time, period hardly changes.

[Total: 9]

4 (a) (i) A cubic container of sides s and cross-sectional area A contains a gas made up of a large number of molecules, each with mass m. One of these molecules has an initial velocity c and a velocity component u_x in the x direction, as shown in Fig. 4.1.

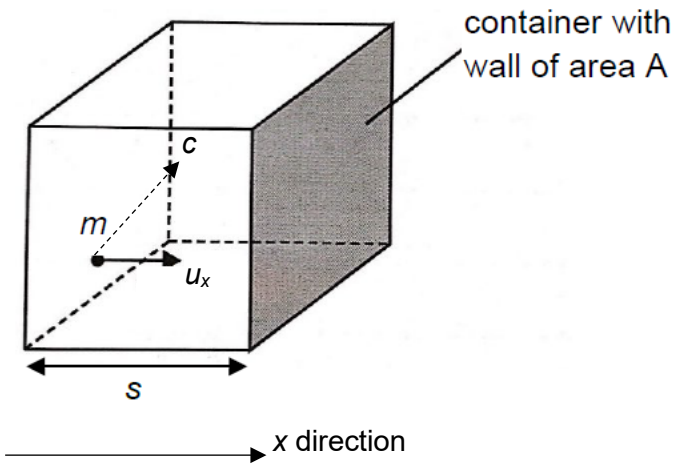


Fig. 4.1

Derive, in terms of m, u_x and s,

1. the change in momentum of the molecule when it collides with the wall on the right, [1]

Change in momentum $\Delta p = m(-u_x) - mu_x = -2mu_x$ [1]

2. the time between consecutive collisions with the wall on the right. [1]

Time between consecutive collisions on the wall, $t = \dfrac{\text{Distance travelled}}{\text{Speed}} = \dfrac{2s}{u_x}$ [1]

(ii) Hence show that the average force F exerted by the molecule on the wall on the right is given by $F = \dfrac{mu_x^2}{s}$. [1]

By Newton's Second Law, average force on particle, $F_p = \dfrac{\Delta p}{t} = \dfrac{-2mu}{2s/u} = -\dfrac{mu^2}{s}$

By Newton's Third Law, average force on wall, $F_w = -F_p = \dfrac{mu_x^2}{s}$ [1]

(iii) Each molecule of the gas has its own velocity c and velocity component u_x in the x direction.

1. State the equation that relates the mean square speed of the molecules, $<c^2>$, to the mean square speed due their velocity components in the x direction, $<u_x^2>$. [1]

As the N particles are in *random* motion, $\langle u_x^{\,2}\rangle = \frac{1}{3}\langle c^2\rangle$ [1]

2. Using your answers to **(a)(ii) and (a)(iii)1**, derive the expression $pV = \frac{1}{3}Nm\langle c^2\rangle$, where p is the pressure on the wall, V is the volume of the container, N is the number of gas particles and $\langle c^2\rangle$ is the mean square speed of the molecules. [2]

Total force of N particles on the wall, $F = N\frac{m\langle u_x^{\,2}\rangle}{s}$ [1]

where $\langle u_x^{\,2}\rangle$ is the mean square speed of the particles.

Pressure on the wall of the container, $p = \frac{F}{s^2} = \frac{Nm\langle u_x^{\,2}\rangle}{s^3} = \frac{Nm\langle u_x^{\,2}\rangle}{V}$

$$p = \frac{1}{3}\frac{Nm\langle c^2\rangle}{V} \quad [1]$$

$$\Rightarrow pV = \frac{1}{3}Nm\langle c^2\rangle$$

(b) A cylinder, shown in Fig. 4.2, fitted with a piston contains ideal gas at a constant temperature of 20 °C. When the volume V is 0.50×10^{-3} m^3, the pressure p in the cylinder is 20×10^4 Pa.

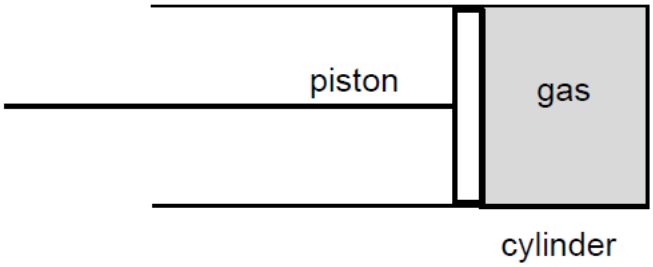


Fig. 4.2

An experiment was conducted and the data was plotted in Fig. 4.3

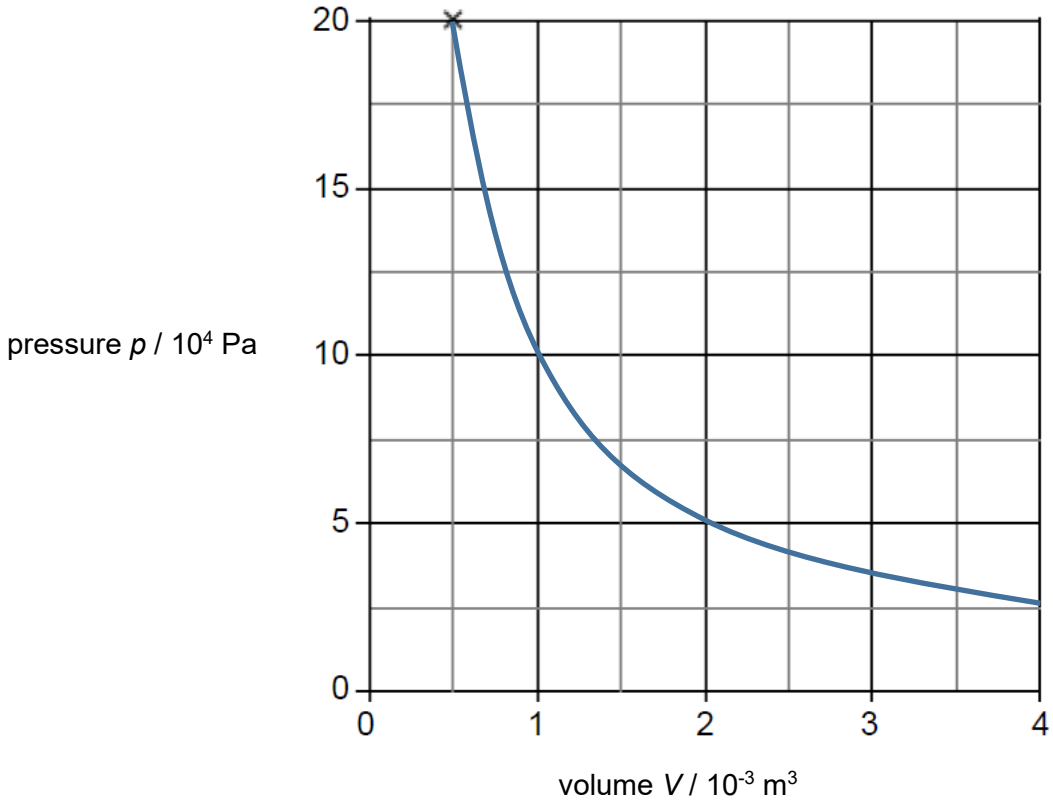


Fig. 4.3

(i) By plotting two additional points, draw a graph on Fig. 4.3 to show the relationship between p and V as the piston is slowly pulled out such that the temperature of the gas remains constant. [1]

(ii) Calculate the number of gas molecules in the cylinder.

Using the ideal gas equation, $pV = NkT$ [1]

$$\text{Number of gas molecules, } N = \frac{pV}{kT} = \frac{\left(20 \times 10^4\right)\left(0.5 \times 10^{-3}\right)}{\left(1.38 \times 10^{-23}\right)\left(20 + 273\right)} = 2.47 \times 10^{22} \quad [1]$$

number of gas molecules = ... [2]

(iii) Determine the total kinetic energy of the gas molecules.

Total kinetic energy, $KE_T = \frac{3}{2}NkT$

$= \frac{3}{2}pV$

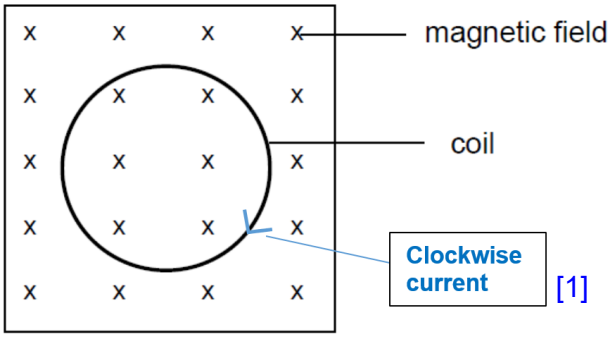
$= \frac{3}{2}\left(20 \times 10^4\right)\left(0.5 \times 10^{-3}\right)$

$= 150\,\text{J}$ [1]

total kinetic energy = ... J [1]

[Total: 10]

5 A coil is placed in a region of magnetic field which is directed into the plane of the paper as shown in Fig. 5.1.



[1]

Fig. 5.1

(a) Define *magnetic flux*.

The magnetic flux through a surface is the product of the component of the magnetic flux density normal to the surface and the area of the surface [1].

[1]

(b) **(i)** The magnitude of the flux density of the magnetic field is decreasing with respect to time. Explain, using Faraday's law, why there is an induced *current* in the coil.

Since the flux density B decreases, then flux linkage Φ = BA decreases. [1]

By Faraday's Law, there is an induced emf in the coil. Since the coil is a closed circuit, the induced emf produces an induced current. [1] [2]

(ii) Draw, on Fig. 5.1, the direction of the induced current. Explain, using Lenz's law, why the induced current is in this direction.

By Lenz's Law, the direction of the induced current must be such as to produce a magnetic flux into the page to *oppose* the decrease of the magnetic flux linkage. [1] Therefore, by the Right Hand Grip Rule, the current will be in the *clockwise* direction to produce an induced magnetic field in the same direction. [2]

[1 for correct direction of induced current]

(c) A magnet of length *x* and magnetic flux density *B* is dropped vertically from rest with the north pole facing downwards. It is dropped from a reference line which is at a vertical height *h* above the top of a *long* solenoid of length *L* and uniform cross-sectional area as shown in Fig. 5.2. The length of the magnet *x* is much smaller than the length of the solenoid *L*.

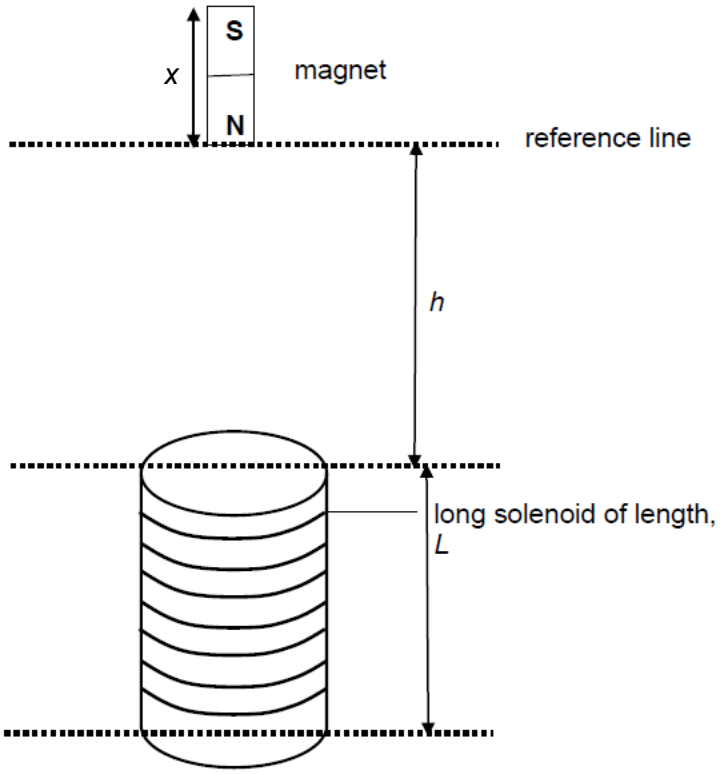


Fig. 5.2

(i) At time t = 0.0 s, the magnet is released from the reference line and allowed to pass through the entire length of the solenoid. Sketch, on Fig. 5.3, a graph to show the variation of the induced e.m.f E in the solenoid with time t. Label this graph as A. (Numerical values are not required and ignore the effects of air resistance.)

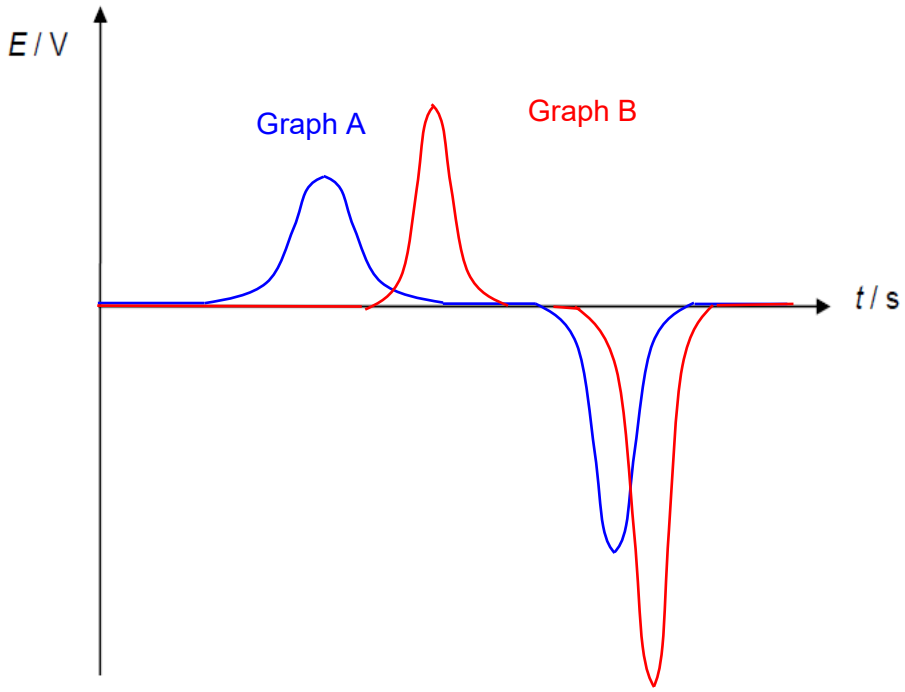


Fig. 5.3

[2]

(ii) State how your graph in part **(c)(i)** would change if the same magnet is released with the south pole facing downwards instead.

The new graph will look like a reflection of Graph A about the time axis [1].

[1]

(iii) The experiment is now repeated with the north pole of the magnet facing downwards, but released from a greater height above the solenoid. Sketch, on Fig. 5.3, a new graph to show the variation of the induced emf E in the solenoid with time t. Label this graph as B.

Both peaks are higher and narrower [1]

Both peaks occur at later time [1]

[2]

[Total: 10]

6 **(a)** A student claims that "Light is only a wave and not a particle."

Discuss whether this statement is correct, providing experimental evidence.

When light passes through two closely spaced slits, bright and dark fringes show the constructive and destructive interference of the light waves. In this case light behaves as waves. [1]

When light above a certain threshold frequency is shone on a metal surface, electrons are emitted from the metal surface. This provides evidence that supports the particulate nature of light. [1]

Therefore, the statement is incorrect. Light is both a wave and a particle. [1]

[3]

(b) A cool, low-pressure gas of an element X is illuminated by a beam of monochromatic violet light of frequency 8.21×10^{14} Hz.

The simplified energy level diagram for an atom of element X is shown in Fig. 6.1 below.

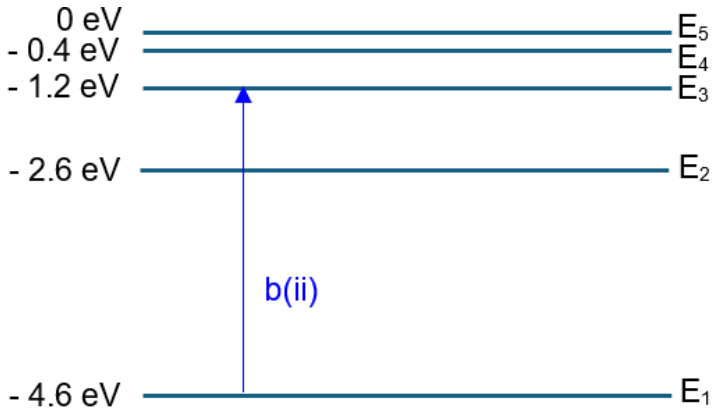


Fig. 6.1

(i) Calculate the energy of a photon of the incident violet light.

$E = hf$

$E = (6.63 \times 10^{-34})(8.21 \times 10^{14}) / 1.6 \times 10^{-19}$ [1]

$E = 3.4$ eV [1]

Energy of a photon = ... eV [2]

(ii) Draw, on Fig. 6.1, the transition corresponding to the absorption of the incident violet photon. [1]

(iii) Determine the number of different emission lines that may be observed as a result of the absorption of many of such violet photons.

Number of emission lines = 3 [1]

(iv) Explain why the intensity of the violet light decreases after passing through the gas.

As the violet light photons are absorbed by the atoms in the cool gas, exciting the electrons from ground state to the higher energy level E_3. [1]

When these excited electrons de-excite, photons of different energies are emitted in **all random directions** rather than continuing purely along the original direction of the beam. The number of violet photons which travel straight through is lower than before, [1] hence lower intensity of violet light is observed. [2]

[Total: 9]

7 Read the passage below and answer the questions that follow.

The Large Hadron Collider (LHC) is the world's largest and most powerful particle accelerator. It first started up on 10 September 2008 and is the crown jewel within the accelerator complex of the European Council for Nuclear Research (CERN).

Spanning across the Switzerland-France border, the LHC is a large horizontal ring in which high-energy particle beams, most commonly proton beams, are accelerated to speeds close to the speed of light in opposite directions. These proton beams circulate around the ring more than 11000 times a second as they are made to collide multiple times at different interaction points where the beams cross each other. By studying the fundamental particles produced in the aftermath of these proton-proton collisions, scientists hope to gain new insights and answers to some of the most important unsolved mysteries in Physics.

Inside the accelerator, the protons travel around the ring for hours along two separate beam pipes, both of which are kept at an ultrahigh vacuum. As the protons circulate around the LHC, they are repeatedly accelerated by strong electric fields along a short linear accelerator within the ring.

The protons are guided around the accelerator ring along the beam pipes by strong magnetic fields generated by thousands of electromagnets of different types and sizes, all of which need to be constantly kept at extremely low temperatures of about 1.9 K. This is why the LHC also houses the world's largest cryogenic system, using about 1.0×10^7 kg of liquid nitrogen and 1.3×10^5 kg of liquid helium to maintain the operation of the magnets.

The most common type of these electromagnets is called dipole magnets, which help bend the protons' trajectories so that they travel successfully along the curved beam pipes. These dipole magnets consist of many coils of electric cables placed next to and at many different sections along the beam pipe, each carrying a large electric current. The dipole magnets are contained and fastened together with strong non-magnetic stainless-steel collars.

Fig. 7.1 shows the cross-section of one of the dipole magnet setups at a section of a beam pipe of the LHC. The large number of electric cables are grouped into two double-layered regions around the beam pipe, labelled as region 1 and region 2 in Fig. 7.1. These electric cables are arranged parallel to the beam pipe such that when a current is travelling along these cables, the magnetic flux density produced by these cables at the centre of the beam pipe helps bend the protons' trajectories so that they can travel along the curved beam pipe.

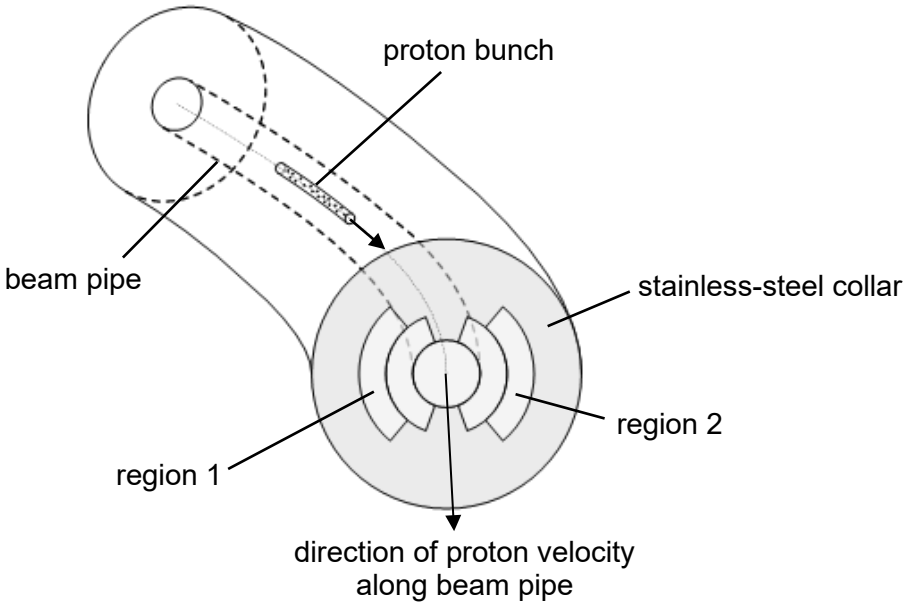


Fig. 7.1
(The electric cables parallel to the beam pipe are not shown for clarity.)

An important parameter in determining the functionality of a particle accelerator like the LHC is the beam current, which is defined as the average rate of flow of charge in the beam pipe. However, during normal operation, the protons do not travel in the beam pipe as a continuous beam. Instead, in each proton beam, the protons travel in groups called "bunches", which are approximately cylindrical in shape as shown in Fig. 7.1.

The LHC is built underground with a mean depth of 100 m for various reasons, including minimising the damage caused to the landscape and environment on the surface. In addition, protons travelling around the curved beam pipes accelerate to produce synchrotron radiation in the form of ultraviolet and X-ray photons, and interactions between the protons and the nuclei of any atoms around the beams in the beam pipes also produce ionising radiation. The Earth's crust therefore also provides natural shielding from these sources of radiation for anyone living on the surface near the LHC.

Table 7.1 shows some important data and parameters of the LHC under normal operating conditions.

Table 7.1

total circumference of LHC ring	26659 m
diameter of beam pipe	56 mm
proton energy	21.2 GeV
current in each cable of a dipole magnet setup	11850 A
length of a dipole magnet setup	14.3 m
peak dipole magnetic flux density at centre of beam pipe	8.33 T
beam current	0.58 A
number of bunches per proton beam	2808
length of one bunch	7.48 cm
cross-sectional area of one bunch	1.0 mm^2

(a) Suggest why the inside of the beam pipe needs to be maintained at an ultrahigh vacuum while the LHC is being operated.

This is to reduce collisions between protons and any gas molecules in the beam pipes, so that less harmful ionising radiation is produced.

OR [1]

This is to reduce collisions between protons and any gas molecules in the beam pipes, so that the proton beam can remain focused and not get scattered by these collisions.

OR

This is to reduce collisions between protons and any gas molecules in the beam pipes, so that there is less energy loss and the protons can reach the high speed required.

(b) State and explain if an accelerator like the LHC can be used to study neutron-neutron collisions.

No. As neutrons are neutral and uncharged, their trajectories cannot be bent / curved by the presence of magnetic fields.

OR [1]

No. As neutrons are neutral and uncharged, they cannot be accelerated by electric fields.

(c) (i) The protons in the LHC beam pipe travel at a speed of 0.999*c*, where *c* is the speed of light. **S**how that the period of revolution of the protons around the LHC ring is 8.89×10^{-5} s.

$$\text{Period } T = \frac{\text{distance travelled along the ring}}{\text{speed of protons}}$$
$$= \frac{26659}{0.999\,c}$$
$$= 8.8863 \times 10^{-5} \text{ s} \quad [1]$$
$$\approx 8.89 \times 10^{-5} \text{ s}$$

[1]

(ii) By considering the beam current, show that the total charge in the ring is 5.16 x 10^{-5} C.

Since beam current is defined as the average rate of flow of charge,

$$\text{beam current} = \frac{\text{total charge in the ring}}{\text{period}} = \frac{Q}{T}$$

$$0.58 = \frac{Q}{8.89 \times 10^{-5}}$$
$$Q = 5.156 \times 10^{-5} \text{ C} \quad [1]$$
$$\approx 5.16 \times 10^{-5} \text{ C}$$

[1]

(iii) Use your answer in **(c)(ii)** to determine the number of protons in each bunch within the proton beam.

Number of protons per proton beam,

$$N = \frac{Q}{e} = \frac{5.16 \times 10^{-5}}{1.60 \times 10^{-19}} = 3.225 \times 10^{14} \qquad [1]$$

Since there are 2808 bunches per proton beam,

$$\text{number of protons in each bunch} = \frac{3.225 \times 10^{14}}{2808}$$
$$= 1.15 \times 10^{11} \quad [1]$$

number of protons = ... [2]

(d) Fig. 7.2 shows a close-up of the cross-section of the dipole magnet setup around the beam pipe in Fig. 7.1, where regions 1 and 2 are represented by combined cables 1 and 2 respectively. Each combined cable carries electric current of the same magnitude.

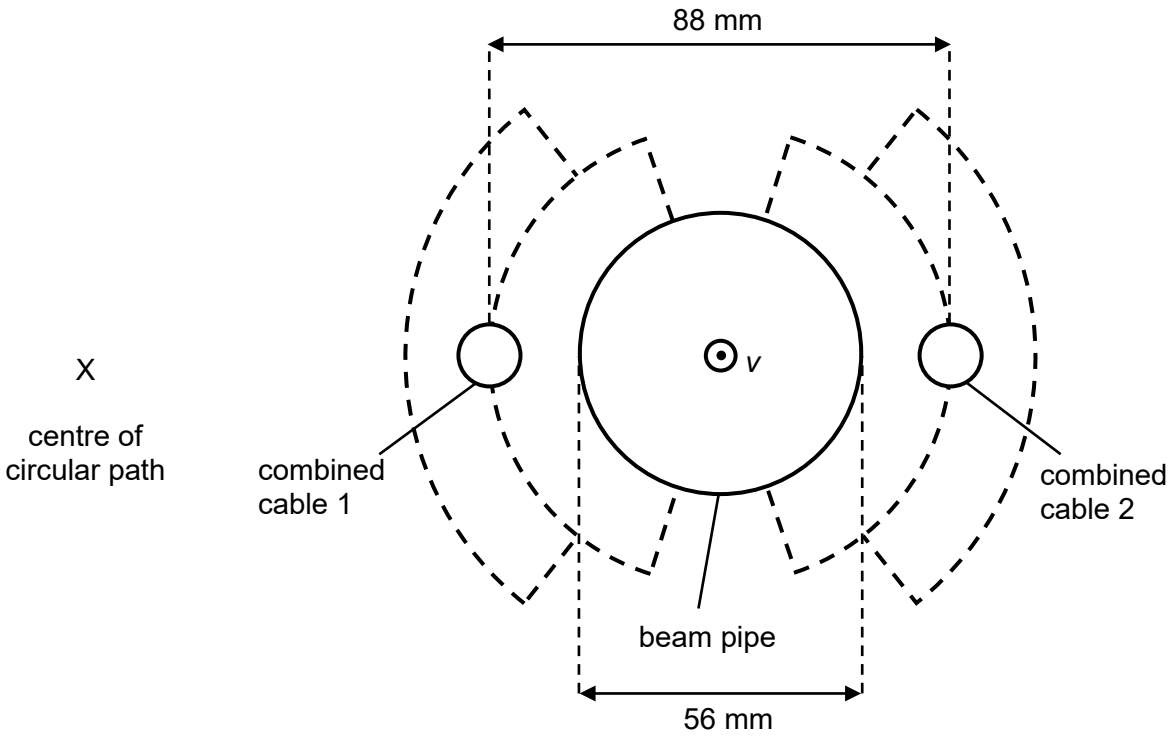


Fig. 7.2

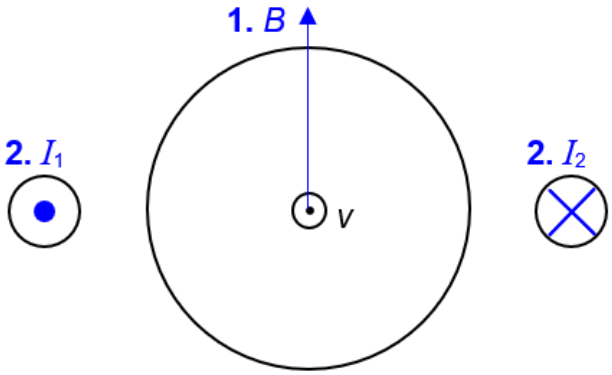
In this model of the dipole magnet setup, the distance between the combined cables is 88 mm, while the diameter of the beam pipe is 56 mm.

The proton velocity *v* in the middle of the beam pipe is pointing out of the page, as shown in Fig. 7.2. The centre of the circular path of the proton's trajectory is on the side of combined cable 1.

(i) On Fig. 7.2, using appropriate symbols and labels, indicate the direction of

1. the magnetic flux density *B* at the centre of the beam pipe,

2. the currents I_1 and I_2 in the combined cables 1 and 2 respectively. [2]



Since proton velocity is out of the page, beam current is out of the page.

Since centre of circular path of the proton's trajectory is on the left, the magnetic force on the proton that provides the centripetal force points to the left.

Using Fleming's left-hand rule, direction of *B* is upwards. [1]

*Straight line with arrowhead pointing upwards at the centre of pipe.

Since each combined cable carries current of the same magnitude, the direction of the current in each cable must be such that each produces a magnetic flux density that points upwards at the centre of the pipe, giving a resultant magnetic flux density as deduced in **(d)(i)1.**

Using the right-hand grip rule, I_1 points out of the page and I_2 points into the page. [1]

*Correct directions, positions and symbols of both currents.

(ii) 1. Determine the current required in each combined cable to produce the peak dipole magnetic flux density at the centre of the beam pipe.

$$B = \frac{\mu_0 I_1}{2\pi d} + \frac{\mu_0 I_2}{2\pi d}$$

$$8.33 = \frac{\left(4\pi \times 10^{-7}\right)\left(I_1 + I_2\right)}{2\pi\left(\frac{88}{2} \times 10^{-3}\right)}$$

$$I_1 + I_2 = 1832600 \text{ A} \quad [1]$$

Since $I_1 = I_2$,

$$I_1 = \frac{1832600}{2}$$

$$= 916300 \text{ A} = 9.16 \times 10^5 \text{ A} \quad [1]$$

current = ... A [2]

2. Hence, determine the number of cables in each region.

Since current in each cable is 11850 A,

$916300 = N(11850)$ where N is the number of cables

$N = 77.3 \approx 77$

number of cables = ... [1]

3. Suggest a reason why the dipole magnets need to be contained and fastened together with collars that are strong.

Support your reason with appropriate calculations.

Force that the combined cables exert on each other,

$$F = B_1 I_2 L$$

$$= \frac{\mu_0 I_1}{2\pi d} I_2 L$$

$$= \frac{\left(4\pi \times 10^{-7}\right)(916300)}{2\pi\left(88 \times 10^{-3}\right)}(916300)(14.3)$$

$$= 2.7287 \times 10^7 = 2.73 \times 10^7 \text{ N} \quad [1]$$

The cables in the dipole magnet setup experience an extremely large repulsive force away from each other. [1] Hence, strong collars are needed to keep the cables from moving away from the beam pipe. [1] .....[3]

(e) Besides dipole magnets, another type of electromagnet used at the LHC are quadrupole magnets.

Quadrupole magnets help "squeeze" the travelling protons within each bunch closer together so that the protons stay travelling along the central axis of the beam pipe in a tightly focused beam. They do so by accounting for the electrostatic repulsion between the protons within each bunch.

(i) By approximating the average volume occupied by a proton in each bunch to be the volume of a cube, show that the average distance between two adjacent protons in a bunch is 8.66×10^{-7} m.

Average volume occupied by one proton

$$= \frac{\text{volume of one bunch}}{\text{no. of protons in one bunch}}$$
$$= \frac{\left(7.48\times10^{-2}\right)\left(1.0\times10^{-6}\right)}{1.15\times10^{11}}$$
$$= 6.5043\times10^{-19}\ \text{m}^3 \qquad [1]$$

Approximating volume occupied by a proton to be volume of a cube with sides of length d:

$$d = \sqrt[3]{6.5043\times10^{-19}}$$
$$= 8.6643\times10^{-7}\ \text{m} \qquad [1]$$
$$\approx 8.66\times10^{-7}\ \text{m} \qquad [2]$$

(ii) Hence determine the magnitude of the average repulsive force between two adjacent protons in a bunch.

Average electric force of repulsion between 2 protons,

$$F_E = \frac{e^2}{4\pi\varepsilon_0 d^2}$$
$$= \frac{\left(1.60\times10^{-19}\right)^2}{4\pi\left(8.85\times10^{-12}\right)\left(8.6643\times10^{-7}\right)^2}$$
$$= 3.0663\times10^{-16} = 3.07\times10^{-16}\ \text{N} \qquad [1]$$

average repulsive force = ... N [1]

(iii) The quadrupole magnets also help to keep the protons in the proton beam on their intended path by accounting for the protons falling downwards due to their weight.

Determine the number of rounds a proton will be able to travel around the LHC accelerator ring before falling to the bottom of the beam pipe due to gravity, if there were no quadrupole magnets.

Assume that the acceleration due to gravity in the pipe is 9.81 m s^{-2}.

Initial velocity v of proton is horizontal along the central axis of the beam pipe.

Displacement of proton from centre of beam pipe to bottom of beam pipe,

$$s_y = u_y t + \frac{1}{2} g t^2$$

$$\frac{1}{2}\left(56 \times 10^{-3}\right) = 0 + \frac{1}{2}(9.81) t^2$$

$t = 0.075554$ s or -0.075554 s (NA) [1]

Total horizontal displacement travelled by proton,

$$s_x = vt$$
$$= (0.999)\left(3.00 \times 10^8\right)(0.075554)$$
$$= 22666199.8 \text{ m} \quad [1]$$

$$\text{Number of rounds} = \frac{s_x}{\text{circumference of ring}}$$
$$= \frac{22666199.8}{26659}$$
$$= 850.23 = 850 \quad [1]$$

number of rounds = ... [3]

[Total: 20]